

Note for section 3

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Abstract

In this note, we discuss various properties of the matrix-product state. We follow Ref. [1,2].

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1 Introduction

Let us consider a collection of L qudits, each of which has a Hilbert space of d -dimension. The full Hilbert space is $\mathcal{H} = \mathcal{H}_{qudit}^{\otimes L}$. A generic wavefunction takes the following form

$$|\Psi\rangle = \sum_{\{\sigma\}} \psi_{\sigma_1\sigma_2\cdots\sigma_L} |\sigma_1\sigma_2\cdots\sigma_L\rangle \quad (1)$$

where $|\sigma\rangle, \sigma = 1, \dots, d$ denote the basis states. Generally, we need d^L complex numbers to specify such a wavefunction. Applying quantum operators on it requires even more resource. Are there better and more efficient ways of representing a wavefunction? In order to do that, let us add more physical ingredients to constrain the wavefunction:

- Symmetry. Usually the state has symmetries. Therefore, we can represent our state within a sub-Hilbert space that has a definite symmetry charge.

For example, if it satisfies a global $\mathbb{Z}_d$ symmetry, we can divide the total Hilbert space into the $\mathbb{Z}_d$ symmetric subspaces and represent our wavefunction only within one subspace, e.g.

$$|\Psi\rangle = \sum_{\mathcal{H}_{\mathbb{Z}_d}} \psi_{\sigma_1\sigma_2\cdots\sigma_L} |\sigma_1\sigma_2\cdots\sigma_L\rangle \quad (2)$$

where $\mathcal{H}_{\mathbb{Z}_d}$ denotes one of the $\mathbb{Z}_d$ symmetric subspace. This can reduce the necessary resource by d times.

Note that this procedure works for mixed states as well.

- Entanglement (locality). If the state is the ground state of a local Hamiltonian, then it generally does not have an extensive amount of entanglement. We can also phrase the statement in a reverse way, a state which always has a large entanglement for any of its subsystem does not have a good notion of locality. This provides us a second way to organize our data.

For example, let us divide our system into A and B with k and $L - k$ qudits respectively. To utilize the entanglement structure, let us write down the Schmidt decomposition

$$|\Psi\rangle = \sum_{\sigma,a} U_{(\sigma_1\cdots\sigma_k),a_\ell} \Lambda_{a_\ell,a_\ell} V_{a_\ell,(\sigma_{k+1}\cdots\sigma_L)} |\sigma_1\cdots\sigma_k\rangle_A \otimes |\sigma_{k+1}\cdots\sigma_L\rangle_B \quad (3)$$

Suppose our state is such that the entanglement between A and B is only $\log 2$.¹ Then it is most likely that Λ only has few large diagonal elements. For simplicity, we assume the simplest scenario $\text{diag}\Lambda = (\frac{1}{2}, \frac{1}{2}, 0, \dots)$, then we only need less than $2(d_A + d_B)$ instead of $d_A d_B$ numbers to specify the state!²

For a mixed state, the Schmidt decomposition fails and we need to look for other approaches.

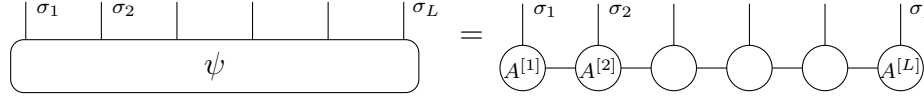
¹the exact number is not important, rather we only require it not to scale with the system size L

²If we further notice that U and V satisfy $U^\dagger U = \mathbb{I}$, $V V^\dagger = \mathbb{I}$ after the truncation, we can further reduce the required amount of resource.

The matrix product state (MPS) is a systematic way to organize the wavefunction via its entanglement structure and is extremely sufficient in one-dimension. The MPS representation of our generic wavefunction is

$$|\Psi\rangle = \sum_{\{\sigma\}} A^{[1]\sigma_1} A^{[2]\sigma_2} \dots A^{[L]\sigma_L} |\sigma_1 \sigma_2 \dots \sigma_L\rangle \quad (4)$$

where $A^{[k]\sigma_k}$ is a $\chi_{k-1} \times \chi_k$ matrix. At the two boundaries, we have $\chi_0 = \chi_L = 1$ and thus $A^{[1]\sigma_1}$ and $A^{[L]\sigma_L}$ are vectors. We will call the indices of matrices as the “bond” indices, and the label of matrices σ as the physical indices. Eq. (4) also has a diagrammatic representation that is usually more convenient to work with



$$\quad (5)$$

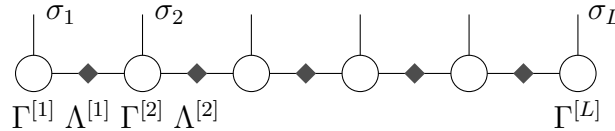
Eq. (4) also implies that the MPS representation has a gauge redundancy, namely, we can locally redefine A by some square matrix X

$$A^{[k]\sigma_k} \rightarrow A^{[k]\sigma_k} X_k, \quad A^{[k+1]\sigma_{k+1}} \rightarrow X_k^{-1} A^{[k+1]\sigma_{k+1}} \quad (6)$$

without changing the actual wavefunction. Another canonical form that makes a more direct contact to Schmidt decomposition is

$$|\Psi\rangle = \sum_{\{\sigma\}} \Gamma^{[1]\sigma_1} \Lambda^{[1]} \Gamma^{[2]\sigma_2} \Lambda^{[2]} \dots \Gamma^{\sigma_{L-1}} \Lambda^{[L-1]} \Gamma^{[L]\sigma_L} |\sigma_1 \dots \sigma_L\rangle \quad (7)$$

where Λ^k is diagonal and encodes the Schmidt coefficients when we bipartite the system at the bond $(k, k+1)$. A diagrammatic representation is the following



$$\quad (8)$$

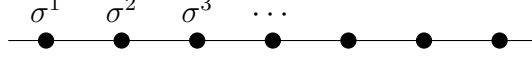
In the rest of this note, we will discuss how to obtain this MPS representations, their properties, evaluation of correlation function. We will also use concrete examples to illustrate the general ideas.

2 Matrix product state

In this section, we introduce various definition of the matrix product state, the left and right-canonical MPS, the canonical MPS and so on. Then we discussion the evaluation of correlation functions. Finally, we use the ground state of the AKLT chain as a concrete example to illustrate the general ideas.

2.1 Definitions, canonical forms

In this subsection, we show that any generic wavefunction (1) can be written in an MPS form. The essential ingredient is the Schmidt decomposition or singular value decomposition.³ There are various equivalent way of doing it, and we will present three of them. In the following discussion, we imagine that the L qudits are arranged on a one-dimensional chain as depicted below so that it is meaningful to talk about left and right.



Left-canonical MPS In this approach, the SVD starts from the left end. In the first step, we reshape the wavefunction into a matrix of dimension $d \times d^{L-1}$

$$\psi_{\sigma_1, (\sigma_2 \cdots \sigma_L)} \equiv \psi_{\sigma_1 \sigma_2 \cdots \sigma_L}$$

where $(\sigma_2 \cdots \sigma_L)$ is regarded as a collective index. We can perform an SVD and have

$$\begin{aligned} \psi_{\sigma_1, (\sigma_2 \cdots \sigma_L)} &= \sum_{a_1} U_{\sigma_1, a_1} \Lambda_{a_1, a_1} (V^\dagger)_{a_1, (\sigma_2 \cdots \sigma_L)} \\ &= \sum_{a_1} A_{a_1}^{\sigma_1} \psi_{a_1 \sigma_2 \cdots \sigma_L} \end{aligned}$$

On the second line, we rename U as A , rename $\Lambda V^\dagger$ as a new wavefunction only for site $2, \cdots, L$. In the next step, we reshape $\psi_{a_1 \sigma_2 \cdots \sigma_L}$ into a matrix $\psi_{(a_1 \sigma_2), (\sigma_3 \cdots \sigma_L)}$ and apply SVD again

$$\begin{aligned} \psi_{(a_1 \sigma_2), (\sigma_3 \cdots \sigma_L)} &= \sum_{a_2} U_{a_1 \sigma_2, a_2} \Lambda_{a_2, a_2} (V^\dagger)_{a_2, (\sigma_3 \cdots \sigma_L)} \\ &= \sum_{a_2} A_{a_1 a_2}^{\sigma_2} \psi_{a_2 \sigma_3 \cdots \sigma_L} \end{aligned}$$

where we apply our naming system again. Note that we always put the physical index σ_i of A upstairs in order to distinguish it from the internal (auxiliary) index a_i , which are put downstairs. We can repeat this procedure until the last site, for which we have

$$\begin{aligned} \psi_{(a_{L-2} \sigma_{L-1}), \sigma_L} &= \sum_{a_{L-1}} U_{(a_{L-2} \sigma_{L-1}), a_{L-1}} \Lambda_{a_{L-1}, a_{L-1}} (V^\dagger)_{a_{L-1}, \sigma_L} \\ &= \sum_{a_{L-1}} A_{a_{L-2} a_{L-1}}^{\sigma_{L-1}} A_{a_{L-1}}^{\sigma_L} \end{aligned}$$

Note the special definition of the last matrix A^{σ_L} , which is actually a vector. Therefore, we can write the original wavefunction (1) as follows

$$|\Psi\rangle = \sum_{\{\sigma\}} A^{\sigma_1} A^{\sigma_2} \cdots A^{\sigma_L} |\sigma_1 \sigma_2 \cdots \sigma_L\rangle \quad (10)$$

³Singular value decomposition. For any $n \times m$ matrix M there is always an SVD

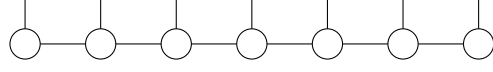
$$M = U \Lambda V^\dagger \quad (9)$$

where U and V are $n \times n$ and $m \times m$ unitary matrices and Λ is a diagonal matrix, whose diagonal elements are non-negative.

or equivalently in terms of the wavefunction components

$$\psi_{\sigma_1 \sigma_2 \dots \sigma_L} = \sum_{a_1, a_2, \dots, a_{L-1}} A_{1, a_1}^{\sigma_1} A_{a_1, a_2}^{\sigma_2} \dots A_{a_{L-2}, a_{L-1}}^{\sigma_{L-1}} A_{a_{L-1}, 1}^{\sigma_L} \quad (11)$$

This is called the left-canonical MPS and graphically represented by


(12)

This fits into the general form (4) we write down before. Generally the dimension of A^{σ_i} will keep increasing and reaches the maximal at $i = L/2$.

The way that we construct the left-canonical MPS is deterministic and free of ambiguity. However, this is not the only way. One can also choose QR decomposition to construct another left-canonical MPS. Physically, they are equivalent. Numerically, QR decomposition is faster but does not give the spectrum of the singular values.

Note that A^σ is the unitary matrix U in each SVD, it satisfies the so-called *left-normalized condition*

$$\sum_{\sigma} (A^\sigma)^\dagger A^\sigma = \mathbb{I} \quad (13)$$

One may worry that this condition can be violated on the last sight, because A^{σ_L} is defined differently. Actually, it also satisfies (13) as long as $|\Psi\rangle$ is well normalized. On the contrary, we generally have

$$\sum_{\sigma} A^\sigma (A^\sigma)^\dagger \neq \mathbb{I}$$

It has the following consequence. Let us group the first k qudits as our subsystem A and the other $L - k$ qudits as our subsystem B , then we rewrite the wavefunction as a sum of bipartite forms

$$|\Psi\rangle = \sum_{a_k} |a_k\rangle_A |a_k\rangle_B \quad (14)$$

$$|a_k\rangle_A = \sum_{\{\sigma\}} (A^{\sigma_1} A^{\sigma_2} \dots A^{\sigma_k})_{1, a_k} |\sigma_1 \dots \sigma_k\rangle, \quad |a_k\rangle_B = \sum_{\{\sigma\}} (A^{\sigma_{k+1}} \dots A^{\sigma_L})_{a_k, 1} |\sigma_{k+1} \dots \sigma_L\rangle$$

One can check that $|a_k\rangle_A$ are orthonormal while $|a_k\rangle_B$ are not.

Right-canonical MPS There is nothing specific in starting SVD from the left. Similarly, we can start from the right and preserve each $V^\dagger$ as the matrix B^σ . This yields the right-canonical MPS

$$|\Psi\rangle = \sum_{\{\sigma\}} B_{1, a_1}^{\sigma_1} \dots B_{a_{L-1}, 1}^{\sigma_{L-1}} |\sigma_1 \sigma_2 \dots \sigma_L\rangle \quad (15)$$

If we bipartite the system, the right components are orthonormal.

Mixed-canonical MPS As the name suggests, this is a mixture of the left and right-canonical MPS. We start SVD from the first site (left) and stop at the k -th site, then the wavefunction becomes

$$\begin{aligned}\psi_{\sigma_1\sigma_2\cdots\sigma_L} &= \sum_{a_k} (A^{\sigma_1} A^{\sigma_2} \cdots A^{\sigma_k})_{1,a_k} \Lambda_{a_k,a_k} (V^\dagger)_{a_k,\sigma_{k+1}\cdots\sigma_L} \\ &= \sum_{a_k} (A^{\sigma_1} A^{\sigma_2} \cdots A^{\sigma_k})_{1,a_k} \psi_{a_k\sigma_{k+1}\cdots\sigma_L}\end{aligned}$$

Then, we do SVD from the L -th site (right) and have

$$\psi_{a_k\sigma_{k+1}\cdots\sigma_L} = \sum U_{a_k\sigma_{k+1},a_{k+1}} \Lambda_{a_{k+1},a_{k+1}} B_{a_{k+1},a_{k+2}}^{\sigma_{k+2}} \cdots B_{a_{L-1},1}^{\sigma_L} \quad (16)$$

or equivalently

$$|\Psi\rangle = \sum_{\sigma} A^{\sigma_1} A^{\sigma_2} \cdots A^{\sigma_k} \Lambda B^{\sigma_{k+1}} B^{\sigma_{k+2}} \cdots B^{\sigma_L} |\sigma_1\sigma_2\cdots\sigma_L\rangle \quad (17)$$

By construction, Λ is the Schmidt coefficient if we bipartite the system at bond $(k, k+1)$.

Canonical MPS So far we have explored three different MPS. In view of the DMRG, it will be useful to have a representation, with which we are able to access the Schmidt coefficients of all $L-1$ possible bipartitionings of the system simultaneously.

Such a notation has been introduced by Vidal and takes the following form

$$|\Psi\rangle = \sum \Gamma^{\sigma_1} \Lambda^{[1]} \Gamma^{\sigma_2} \Lambda^{[2]} \cdots \Gamma^{\sigma_{L-1}} \Lambda^{[L-1]} \Gamma^{\sigma_L} |\sigma_1 \cdots \sigma_L\rangle \quad (18)$$

where Γ 's live on the sites and Λ 's live on the bonds. Furthermore, $\Lambda^{[k]}$ is a diagonal matrix and records the Schmidt coefficients if we cut our system at the bond $(k, k+1)$. Namely, the Schmidt decomposition for the bipartition at bond $(k, k+1)$ is

$$|\Psi\rangle = \sum_{a_k} \Lambda_{a_k}^{[k]} |a_k\rangle_A |a_k\rangle_B \quad (19)$$

where

$$\begin{aligned}|a_k\rangle_A &= \sum_{\{\sigma\}} (\Gamma^{\sigma_1} \Lambda^{[1]} \cdots \Lambda^{[k-1]} \Gamma^{\sigma_k})_{1,a_k} |\sigma_1 \cdots \sigma_k\rangle, \\ |a_k\rangle_B &= \sum_{\{\sigma\}} (\Gamma^{\sigma_{k+1}} \Lambda^{[k+1]} \cdots \Lambda^{[L-1]} \Gamma^{\sigma_L})_{a_k,\sigma_{k+1}\cdots\sigma_L} |\sigma_{k+1} \cdots \sigma_L\rangle\end{aligned}$$

are orthonormal.

Let us show that the generic wavefunction (1) can indeed be brought into the form (18) in a procedure that is analogous to the one that produces the left-canonical MPS. The first step is the same as before

$$\begin{aligned}\psi_{\sigma_1\cdots\sigma_L} &= \psi_{\sigma_1,(\sigma_2\cdots\sigma_L)} \\ &= \sum_{a_1} \Gamma_{a_1}^{\sigma_1} \Lambda_{a_1,a_1}^{[1]} (V^\dagger)_{a_1,\sigma_2\cdots\sigma_L}\end{aligned}$$

In the next step, we apply SVD again to the wavefunction $\psi_{a_1\sigma^2\cdots\sigma^L} = \Lambda_{a_1,a_1}^{[1]}(V^\dagger)_{a_1,\sigma^2\cdots\sigma^L}$, but we will use the singular value matrix $\Lambda^{[1]}$ obtained before to write the result as

$$\begin{aligned}\psi_{a_1\sigma^2\cdots\sigma^L} &= \sum_{a_2} U_{a_1,a_2}^{\sigma^2} \Lambda_{a_2,a_2}^{[2]} (V^\dagger)_{a_2,\sigma^3\cdots\sigma^L} \\ &= \sum_{a_1} \Lambda_{a_1,a_1}^{[1]} \Gamma_{a_1,a_2}^{[\sigma^2]} \Lambda_{a_2,a_2}^{[2]} (V^\dagger)_{a_2,\sigma^3\cdots\sigma^L} \\ &= \sum_{a_1} \Lambda_{a_1,a_1}^{[1]} \Gamma_{a_1,a_2}^{[\sigma^2]} \psi_{a_2\sigma^3,\sigma^4\cdots\sigma^L}\end{aligned}$$

Repeating the steps produces (18). By construction, the left-hand side of $\Lambda^{[k]}$ is product of U 's, the right-hand side is $V^\dagger$. This gives the orthonormal condition we want.

2.2 Correlation functions of MPS

In this section, we discuss the evaluation of correlation functions of MPS. We will assume that our MPS describes a quasi-one-dimensional system.

Results in the previous section are completely general and useful in certain numerical calculation such as DMRG and TEBD. In order to develop better analytical understanding of the MPS wavefunction, it is convenient to consider more specific and simpler MPS wavefunctions. For example, we are usually interested in states defined on a closed chain that preserve translation symmetry.⁴ Then, it is intuitive to assume that such a state is described by the following *translationally invariant MPS*

$$|\Psi\rangle = \sum_{\{\sigma\}} \text{Tr} (A^{\sigma_1} A^{\sigma_2} \cdots A^{\sigma_L}) |\sigma_1 \sigma_2 \cdots \sigma_L\rangle \quad (20)$$

in which different sites are assigned with the same set of matrices. The appearance of trace is because of the periodic boundary condition. In the presence of boundaries, we need additional data to specify the boundary degrees of freedom. In the simplest case, the boundaries can be captured by two vectors v_L and v_R to write the MPS as

$$|\Psi_{v_L,v_R}\rangle = \sum_{\{\sigma\}} v_L^\dagger A^{\sigma_1} A^{\sigma_2} \cdots A^{\sigma_L} v_R |\sigma_1 \sigma_2 \cdots \sigma_L\rangle \quad (21)$$

In the following, we will consider the correlation functions of the translationally invariant MPS (20). We can also regard it as the correlators defined deep inside the bulk of the MPS with open boundary condition (21).

We will see later that it is convenient to introduce an auxiliary quantity, called the double tensor

$$E^{[k]} = \sum_{\sigma_k} A^{\sigma_k,*} \otimes A^{\sigma_k} = \begin{array}{c} A^* \\ \vdots \\ A \end{array} \quad (22)$$

⁴Even when there is symmetry breaking, we can still enlarge our unit cell so that there is still a translation symmetry for the enlarged unit cell.

which can also be regarded as a matrix with collective indices. It becomes Hermitian if we collect the indices as

$$E_{\alpha\beta,\gamma\delta}^{[k]} = \sum_{\sigma_k} A_{\alpha,\beta}^{\sigma_k,*} A_{\gamma,\delta}^{\sigma_k} \quad (23)$$

In the evaluation of the correlation functions, it is also useful to group the indices as

$$E_{\alpha\gamma,\beta\delta}^{[k]} = \sum_{\sigma_k} A_{\alpha,\beta}^{\sigma_k,*} A_{\gamma,\delta}^{\sigma_k} = \begin{array}{c} \alpha \text{---} \bullet \text{---} \beta \\ \quad \quad \quad \uparrow A^* \\ \gamma \text{---} \bullet \text{---} \delta \\ \quad \quad \quad \downarrow A \end{array} \quad (24)$$

For a translationally invariant MPS $|\Psi\rangle$ with periodic boundary condition, the norm of a state is⁵

$$\langle\Psi|\Psi\rangle = \text{Tr } E^L = \sum_n \lambda_n^L \quad (26)$$

where λ_n are the eigenvalues of E , and they are usually ordered as $\lambda_1 \geq \lambda_2 \geq \dots$.

The expectation value of an operator that acts on a single site k is

$$\langle O \rangle = \frac{\text{Tr}(E[O]E^{L-1})}{\text{Tr } E^L} \quad (27)$$

where we have defined

$$E[O] \equiv \sum_{\sigma_k, \sigma'_k} O_{\sigma_k, \sigma'_k} M^{\sigma_k,*} \otimes M^{\sigma'_k} \quad (28)$$

We can introduce the eigenvectors $|v_n\rangle$ of E and have

$$\langle O \rangle = \frac{1}{\sum_n \lambda_n^L} \sum_n \langle v_n | E[O] | v_n \rangle \lambda_n^{L-1} \quad (29)$$

In the thermodynamic limit, we only need to preserve the largest eigenvalues and the result simplifies to

$$\langle O \rangle = \frac{1}{\lambda_1} \sum_{n=1}^{deg} \langle v_n | E[O] | v_n \rangle, \quad L \gg 1 \quad (30)$$

where $deg \geq 1$ denotes the degeneracy of the largest eigenvalues.

Similarly, the two-point function of a local operator is

$$\begin{aligned} \langle O_j O_k \rangle &= \frac{\text{Tr}(E[O_k]E^{j-k-1}E[O_j]E^{L-j+k-1})}{\text{Tr } E^L} \\ &= \frac{1}{\sum_n \lambda_n^L} \sum_{n,m} \langle v_n | E[O] | v_m \rangle \langle v_m | E[O] | v_n \rangle \lambda_m^{j-k-1} \lambda_n^{L-j+k-1} \end{aligned} \quad (31)$$

⁵Without translation invariance, the norm of a state is

$$\langle\psi|\psi\rangle = \text{Tr} \left(E^{[1]} E^{[2]} \dots E^{[L]} \right) \quad (25)$$

where we rewrite the result using the eigenvectors of E . In the thermodynamic limit, we only need preserve the largest eigenvalues for λ_n and have

$$\begin{aligned} \langle O_j O_k \rangle = & \frac{1}{\lambda_1^2} \left(\sum_{n=1}^{deg} \langle v_n | E[O] | v_n \rangle \right)^2 + \frac{1}{\lambda_1^2} \sum_{n \neq m}^{deg} \langle v_n | E[O] | v_m \rangle \langle v_m | E[O] | v_n \rangle \\ & + \frac{1}{\lambda_1^2} \sum_{n=1}^{deg} \sum_{m > deg} \left(\frac{\lambda_m}{\lambda_n} \right)^{j-x-1} \langle v_n | E[O] | v_m \rangle \langle v_m | E[O] | v_n \rangle \quad L \gg 1 \end{aligned} \quad (32)$$

Note that we have separate terms with different physical meanings:

- The first term is the disconnect two-point function.
- The second term does not decay even at long distance and only appears when the largest eigenvalue of E has degeneracy. Physically, it implies symmetry breaking and the state $|\Psi\rangle$ is in a macroscopic cat state.
- The last term exponentially decays with the distance. The correlation length is controlled by the gap in the spectrum of E .

3 Example: AKLT ground state

In this section, we consider a concrete example, the ground state of the AKLT Hamiltonian. It is one of the simplest examples of one-dimensional SPTs. First, we will show how to write down the MPS wavefunction by using the idea of *projected entangled pair state* (PEPS), which also illustrate the physical meaning of bond indices. Then, we will evaluate some correlation functions to extract some of its physical properties. Finally, we will examine how the symmetry acts on the wavefunction.

3.1 Construction of the ground state wavefunction: PEPS

Let us consider a one-dimensional chain of spin-1's. The AKLT Hamiltonian entangles every nearest neighboring spins in a special way

$$H = \sum_i 2P_{i,i+1}^{J=2} - \frac{2}{3} = \frac{1}{2} \sum_i s_i \cdot s_{i+1} + \frac{1}{3} (s_i \cdot s_{i+1})^2 \quad (33)$$

where $(P_{i,i+1}^{J=2})^2 = \mathbb{I}$ is a projection operator, which projects each neighboring spins onto the total spin-2 subspace. It has a lattice translation symmetry, on-site $\text{SO}(3)$ symmetry and time-reversal.⁶

Let us give an explicit construction of the ground state wavefunction. For simplicity, we choose to work with periodic boundary condition. The construction consists of three steps

⁶One may instead claim that the symmetry group is $\text{SU}(2)$ instead of $\text{SO}(3)$. The reason we prefer $\text{SO}(3)$ is that the local operators and thus local excitations (in the bulk) are in the linear representation of $\text{SO}(3)$.

1. Decompose each spin-1 into two spin- $\frac{1}{2}$'s. Let a_k, b_k denotes the two spin- $\frac{1}{2}$ in the k -th unit cell. The Hilbert space is enlarged in this step and we need to finally project the state in the enlarged Hilbert space back into the physical Hilbert space.
2. Let two neighboring spin- $\frac{1}{2}$'s that are in different unit cells form a singlet. For example, for b_k and a_{k+1} , their wavefunction takes the form

$$|\phi\rangle_{k,k+1} = \sum_{b_k, a_{k+1}} \Sigma_{b_k, a_{k+1}} |b_k, a_{k+1}\rangle, \quad \Sigma = \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & 0 \end{pmatrix} \quad (34)$$

Under periodic boundary condition, the full wavefunction becomes

$$|\Psi'\rangle = \sum \Sigma_{b_1, a_2} \Sigma_{b_2, a_3} \cdots \Sigma_{b_L, a_1} |a_1, b_1\rangle \otimes |a_2, b_2\rangle \cdots |a_L, b_L\rangle \quad (35)$$

3. Project the pair of spin- $\frac{1}{2}$'s in the same unit cell onto the spin-1 subspace. Let us introduce the projector

$$P^{J=1} = \sum_{\sigma, a, b} M_{a, b}^{\sigma} |\sigma\rangle \langle ab|, \quad M^+ = \begin{pmatrix} 1 & \\ & 0 \end{pmatrix}, \quad M^0 = \begin{pmatrix} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \end{pmatrix}, \quad M^- = \begin{pmatrix} 0 & \\ & 1 \end{pmatrix} \quad (36)$$

Applying it onto $|\Psi'\rangle$, we get the desired ground state

$$\prod_k P_k^{J=1} |\Psi'\rangle = \sum_{\sigma} \text{Tr} (M^{\sigma_1} \Sigma M^{\sigma_2} \Sigma \cdots M^{\sigma_L} \Sigma) |\sigma_1 \sigma_2 \cdots \sigma_L\rangle \quad (37)$$

where the appearance of Tr is a characteristic of the periodic boundary condition, the wavefunction is also manifestly translationally invariant.

We can further simplify the representation by introducing $\tilde{A}^{\sigma} = M^{\sigma} \Sigma$

$$\tilde{A}^+ = \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} \\ 0 & 0 \end{pmatrix}, \quad \tilde{A}^0 = \begin{pmatrix} -\frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}, \quad \tilde{A}^- = \begin{pmatrix} 0 & 0 \\ -\frac{1}{\sqrt{2}} & 0 \end{pmatrix} \quad (38)$$

and rewrite the wavefunction as

$$|\Psi\rangle = \sum_{\sigma} \text{Tr} (\tilde{A}^{\sigma_1} \tilde{A}^{\sigma_2} \cdots \tilde{A}^{\sigma_L}) |\sigma_1 \sigma_2 \cdots \sigma_L\rangle \quad (39)$$

Note that A^{σ} is a 2×2 matrix, we call that such a wavefunction has a bond dimension equal to 2.

3.2 Physical properties

In the following, let us examine some physical properties of this wavefunction.

Normalization $|\Psi\rangle$ is obtained by performing projection of a normalized wavefunction and thus is not normalized. Indeed, let us check its normalization

$$\begin{aligned}\langle\Psi|\Psi\rangle &= \sum_{\{\sigma\}} \text{Tr} \left(\tilde{A}^{\sigma_1} \tilde{A}^{\sigma_2} \dots \tilde{A}^{\sigma_L} \right)^* \text{Tr} \left(\tilde{A}^{\sigma_1} \tilde{A}^{\sigma_2} \dots \tilde{A}^{\sigma_L} \right) \\ &= \text{Tr} \left[\left(\sum_{\sigma_1} \tilde{A}^{\sigma_1,*} \otimes \tilde{A}^{\sigma_1} \right) \dots \left(\sum_{\sigma_L} \tilde{A}^{\sigma_L,*} \otimes \tilde{A}^{\sigma_L} \right) \right]\end{aligned}\quad (40)$$

Let us follow the general discussion and introduce the double tensor

$$E = \sum_{\sigma} \tilde{A}^{\sigma,*} \otimes \tilde{A}^{\sigma} = \begin{pmatrix} \frac{1}{4} & & \frac{1}{2} \\ & -\frac{1}{4} & \\ \frac{1}{2} & & -\frac{1}{4} \\ & & & \frac{1}{4} \end{pmatrix}\quad (41)$$

which has eigenvalues $\frac{3}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}$. The wavefunction norm can be written in a simple way in terms of E and its eigenvalues λ_n

$$\langle\Psi|\Psi\rangle = \text{Tr} E^L = \sum_{n=1}^4 \lambda_n^L \stackrel{L \gg 1}{\approx} \left(\frac{3}{4}\right)^L\quad (42)$$

So we can rescale $\tilde{A}^{\sigma}$ to $A^{\sigma} = \frac{2}{\sqrt{3}} \tilde{A}^{\sigma}$ to insure the correct normalization. And the ordinary AKLT wavefunction refers to the one written in terms of A^{σ} , i.e.

$$|\Psi_{AKLT}\rangle = \sum_{\sigma} \text{Tr} (A^{\sigma_1} A^{\sigma_2} \dots A^{\sigma_L}) |\sigma_1 \sigma_2 \dots \sigma_L\rangle\quad (43)$$

where the matrices are

$$A^+ = \sqrt{\frac{2}{3}} \frac{\sigma^x + i\sigma^y}{2}, \quad A^0 = -\frac{1}{\sqrt{3}} \sigma^z, \quad A^- = -\sqrt{\frac{2}{3}} \frac{\sigma^x - i\sigma^y}{2}\quad (44)$$

This representation is not unique. We can also rotate the local basis such that the matrices take simpler form

$$A^x = \frac{1}{\sqrt{3}} \sigma^x, \quad A^y = \frac{1}{\sqrt{3}} \sigma^y, \quad A^z = \frac{1}{\sqrt{3}} \sigma^z\quad (45)$$

and the corresponding basis states are the eigenvalue-0 eigenstates of the spin in the x, y and z directions respectively

$$|x\rangle = \frac{|+\rangle - |-\rangle}{\sqrt{2}}, \quad |y\rangle = i \frac{|+\rangle + |-\rangle}{\sqrt{2}}, \quad |z\rangle = -|0\rangle\quad (46)$$

Correlation function Let us begin with the one-point function of a spin operator O_j , where $O_j = s_x, s_y$ or s_z

$$\langle O_j \rangle = \frac{\langle \Psi_{AKLT} | O_j | \Psi_{AKLT} \rangle}{\langle \Psi_{AKLT} | \Psi_{AKLT} \rangle}\quad (47)$$

Using the double tensor E , we can write it as

$$\langle O_j \rangle = \frac{\text{Tr}(E[O]E^{L-1})}{\text{Tr } E^L}, \quad E[O] = \sum_{\sigma, \sigma'} O_{\sigma, \sigma'} A^{\sigma, *} \otimes A^{\sigma'} \quad (48)$$

Let $|v_n\rangle$ denote the eigenvectors of E , then we have

$$\langle O_j \rangle = \frac{\sum_{n=1}^4 \lambda_n^{L-1} \langle v_n | E[O] | v_n \rangle}{\sum_{n=1}^4 \lambda_n^L} \stackrel{L \gg 1}{\approx} \lambda_1^{-1} \langle v_1 | E[O] | v_1 \rangle \quad (49)$$

Let us consider a simple example $O = s_z = \text{diag}(1, 0, -1)$ and have

$$E[O = s_z] = \sum_{\sigma} (s_z)_{\sigma, \sigma} A^{\sigma, *} \otimes A^{\sigma} = \begin{pmatrix} 0 & 0 & 0 & \frac{2}{3} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -\frac{2}{3} & 0 & 0 & 0 \end{pmatrix} \quad (50)$$

Then, its one-point function vanishes, because $v_1 = (\frac{1}{\sqrt{2}}, 0, 0, \frac{1}{\sqrt{2}})^T$.

Let us then consider a two-point function of the local operator O

$$\langle O_0 O_x \rangle = \frac{\langle \Psi_{AKLT} | O_0 O_x | \Psi_{AKLT} \rangle}{\langle \Psi_{AKLT} | \Psi_{AKLT} \rangle} = \frac{\text{Tr}(E[O]E^x E[O]E^{L-x-2})}{\text{Tr } E^L} \quad (51)$$

We can apply the same method by introducing the eigenvectors $|v_n\rangle$ of E and write

$$\langle O_0 O_x \rangle = \frac{1}{\sum_n \lambda_n^L} \sum_{n, m} \lambda_n^{L-2} \left(\frac{\lambda_m}{\lambda_n} \right)^x \langle v_n | E[O] | v_m \rangle \langle v_m | E[O] | v_n \rangle \quad (52)$$

Noting that for the AKLT MPS, the three subleading eigenvalues are degenerate, we have

$$\langle O_0 O_x \rangle = \left(\frac{\langle v_1 | E[O] | v_1 \rangle}{\lambda_1} \right)^2 + \left(\frac{\lambda_2}{\lambda_1} \right)^x \sum_{m=2}^4 \langle v_1 | E[O] | v_m \rangle \langle v_m | E[O] | v_1 \rangle \quad (53)$$

We can recognize that the first term is the disconnected piece, so the connected two-point function is exponentially decaying.

This is a special example, where the largest eigenvalue of E is not degenerate. Let us consider what happens if it has degeneracy. Then it implies long-range order.

Symmetries The Hamiltonian has an $\text{SO}(3)$ symmetry and we want to examine how it acts on our MPS wavefunction. Because the wavefunction has a translational invariance, it suffice to check how it acts locally, namely

$$R(g)_{\sigma\sigma'} A^{\sigma'} \quad (54)$$

where $R(g)$ is the fundamental representation matrix of a group element $g \in \text{SO}(3)$. Note that A^{σ} consists of Pauli matrices which behaves as $\text{SO}(3)$ vectors in the adjoint representation of $\text{SU}(2)$. Thus, we have

$$R(g)_{\sigma\sigma'} A^{\sigma'} = U_g A^{\sigma} U_g^{\dagger} \quad (55)$$

where U_g is a $SU(2)$ matrix that corresponds to $g \in SO(3)$. Namely, the symmetry action can be pushed from physical index to the bond index. In this example, the symmetry action on the bond index is in the projective representation of $SO(3)$. This is a generic feature of the one-dimensional symmetry protected bosonic topological phase.

One of the physical consequences of such a fractionalized symmetry is that we can instead consider the following correlation function

$$\langle s_j^z \prod_{j < \ell < k} e^{i\pi s_\ell^z} s_k^z \rangle \quad (56)$$

which is a two-point function of s^z but dressed by a string of π -rotation. With the insertion of a string, one can verify that it no longer decays with the distance $k - j$.

4 Mean-field theory with MPS

In this section, we will apply the MPS to improve the mean-field theory solution to the one-dimensional transverse-field Ising model (TFIM). For simplicity, we will consider a translationally invariant MPS with bond dimension $\chi = 2$.

Let X, Y, Z denote the Pauli operator and $|0\rangle, |1\rangle$ denote the eigenstates of Z with eigenvalues ± 1 . The Hamiltonian for the TFIM is

$$H = -J \sum_i Z_i Z_{i+1} - h \sum_i X_i \quad (57)$$

In the simplest mean-field approach, the ansatz wavefunction is a local product state with zero correlation and a single variational parameter, which can be written as

$$(\cos \theta |0\rangle + \sin \theta |1\rangle)^{\otimes N} \quad (58)$$

It interpolates between a symmetry breaking state and a symmetric state when θ increases from 0 to $\pi/2$. Now, we want to promote it to a $\chi = 2$ translationally invariant MPS

$$|\Psi\rangle = \sum_{\{\sigma\}} \text{Tr}(A^{\sigma_1} A^{\sigma_2} \cdots A^{\sigma_L}) |\sigma_1 \sigma_2 \cdots \sigma_L\rangle \quad (59)$$

where each A^σ is a 2×2 matrix. The simplest ansatz (58) breaks the $\mathbb{Z}_2$ symmetry explicitly. We will assume that the MPS ansatz (59) obeys the $\mathbb{Z}_2$ symmetry, which will allow us to give strong constraints on it.

4.1 Number of parameters

Naively, it seems that there are eight complex parameters that we can tune. However, it turns out that only two of them are actually physical.⁷ This has to do with the gauge degrees of freedom in the MPS wavefunction, which does

$$A^\sigma \rightarrow M A^\sigma M^{-1}, \quad M \in \text{GL}(2, \mathbb{C}) \quad (60)$$

⁷We thank Yi-Zhuang You for helpful private discussion on a related problem.

This “gauge symmetry” combined with the physical symmetries gives a strong constraint on the number of independent physical parameters. The most general form of A^σ is

$$A^\sigma = c_0^\sigma \mathbb{I} + c_1^\sigma X + c_2^\sigma Y + c_3^\sigma Z \quad (61)$$

where $c_{0,1,2,3}^\sigma$ are complex numbers. Now, let us remove the redundant ones:

1. First of all, we can apply a gauge transformation $A^\sigma \rightarrow M A^\sigma M^{-1}$ to remove the Y term, where $M = a\mathbb{I} + bX$ and $a^2 + b^2 = 2abc_3/c_2$. For the other two parameters c_1, c_3 , their signs can be changed arbitrarily by gauge transformation with X or Z .
2. Second, we require the wavefunction to respect the on-site $\mathbb{Z}_2$ symmetry, which says

$$\exists M \in \text{GL}(2, \mathbb{R}), \quad X_{\sigma, \sigma'} A^{\sigma'} = M A^\sigma M^{-1} \quad (62)$$

If pick the gauge where $M = X$, the matrix A^σ is restricted to

$$A^\sigma = c_0 \mathbb{I} + c_1 X + (-1)^\sigma c_3 Z \quad (63)$$

One can also pick a different gauge where $M = X$, then the matrix takes a slightly different form

$$A^\sigma = c_0 \mathbb{I} + (-1)^\sigma c_1 X + c_3 Z \quad (64)$$

The two gauge are related by an Hadamard gate. It is sometimes also convenient to write the matrix A in the eigenbasis of Pauli- X . In the two different gauges, it takes the following form

$$\begin{aligned} \text{X-gauge:} \quad & A^+ = \sqrt{2} (c_0 \mathbb{I} + c_1 X), \quad A^- = \sqrt{2} c_3 Z \\ \text{Z-gauge:} \quad & A^+ = \sqrt{2} (c_0 \mathbb{I} + c_3 Z), \quad A^- = \sqrt{2} c_1 X \end{aligned} \quad (65)$$

3. Then, note that the Hamiltonian is real, which allows us to write a purely real wavefunction. Thus, we can choose $c_{0,1,3}$ to be all real and further $c_{1,3}$ are positive.
4. Finally, note that we can always rescale the matrix by a constant and take care of the normalization at the end of the calculation. Therefore, we only have two independent of parameters.

This also provides some freedom in the choice of parametrization. For example, one convenient parametrization that will be used for the numerical calculation is

$$A^\sigma = \mathbb{I} + c (\sin \theta X + (-1)^\sigma \cos \theta Z), \quad c > 0, \quad 0 < \theta < \frac{\pi}{2} \quad (66)$$

In the numerics, it is found that only the regime $c \in [0, 1]$ is physically more interesting. In this regime, another form is convenient for understanding the physical meaning of parameters in the ferromagnetic phase

$$A^\sigma = e^{\rho(\sin \theta X + (-1)^\sigma \cos \theta Z)} \quad (67)$$

Equivalently, we have $c_3 < c_0$ in this regime, so we can also parametrize $A^\pm$ (in the Z -gauge) in a similar way and have

$$A^+ = \sqrt{2} e^{\alpha Z}, \quad A^- = \sqrt{2} \beta X \quad (68)$$

which will be useful for understanding the physical meaning of parameters in the paramagnetic phase.

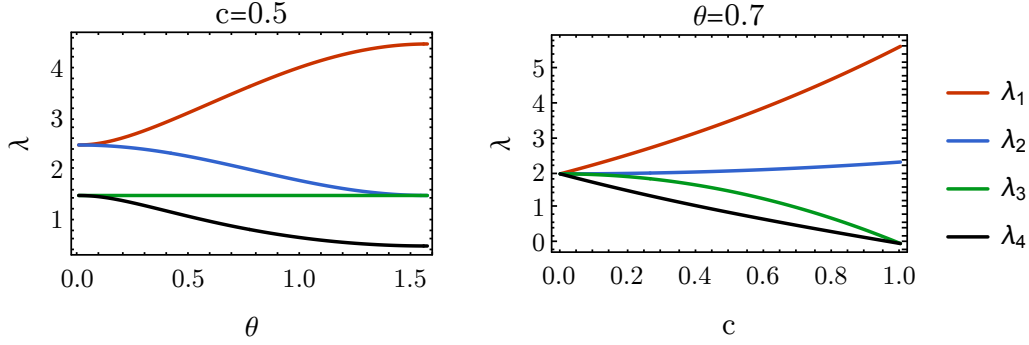


Figure 1: The four eigenvalues of the double tensor E at different values of c and θ .

4.2 Physical meaning of the parameters

In this subsections, we will identify the ferromagnetic and paramagnetic phase in the parameter space. Then, we extract the physical meaning of the parameters, for example, which part describes the domain wall tension and so on. We want to show:

- When $\theta = 0$, the wavefunction is in the ferromagnetic phase, $\tanh^{-1} c$ is the line tension of domain wall.
- When $\theta > 0$, the wavefunction is in the paramagnetic phase, $c \sin \theta$ is the probability of the $\mathbb{Z}_2$ charge creation, $\tanh^{-1}(c \cos \theta)$ is the virtual energy cost for the charge propagation.

The general analysis in Sec.2.2 tells us that the symmetry breaking is associated to the degeneracy of the largest eigenvalue of the double tensor. This provides a simple way to identify the ferromagnetic phase. The double tensor is defined as $E_{\alpha\gamma,\beta\delta} = \sum_{\sigma} A_{\alpha,\beta}^{\sigma} A_{\gamma,\delta}^{\sigma}$ and can be written as a 4×4 matrix in this case

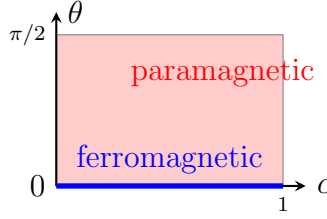
$$E = 2 \begin{pmatrix} 1 + c^2 \cos^2 \theta & c \sin \theta & c \sin \theta & c^2 \sin^2 \theta \\ c \sin \theta & 1 - c^2 \cos^2 \theta & c^2 \sin^2 \theta & c \sin \theta \\ c \sin \theta & c^2 \sin^2 \theta & 1 - c^2 \cos^2 \theta & c \sin \theta \\ c^2 \sin^2 \theta & c \sin \theta & c \sin \theta & 1 + c^2 \cos^2 \theta \end{pmatrix} \quad (69)$$

It generally has four different eigenvalues

$$\begin{aligned} \lambda_1 &= 2 + c^2 - c^2 \cos(2\theta) + c \sqrt{\frac{16 + 3c^2 + 4(c^2 - 4) \cos(2\theta) + c^2 \cos(4\theta)}{2}} \\ \lambda_2 &= 2(1 + c^2 \cos(2\theta)) \\ \lambda_3 &= 2(1 - c^2) \\ \lambda_4 &= 2 + c^2 - c^2 \cos(2\theta) - c \sqrt{\frac{16 + 3c^2 + 4(c^2 - 4) \cos(2\theta) + c^2 \cos(4\theta)}{2}} \end{aligned} \quad (70)$$

Their values at general c and θ are plotted in Fig.1 and we can see their relative ordering $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4$. The largest eigenvalue have a two-fold degeneracy at $\theta = 0$, which means that the ferromagnetic phase is on the line $\theta = 0$ and the paramagnetic phase occupies the rest

of the parameter space, i.e.



Now, let us figure out the physical meaning of each parameter in two phases. We will switch between different gauges and parameterization to simplify the analysis. In the end we will translate the conclusion to the (c, θ) representation.

- In the ferromagnetic phase, domain walls have a extensive energy cost and are strongly suppressed. Namely, the weight of a domain wall configuration in the ground state wavefunction is exponentially small in the length of the domain wall. Now, we want to understand how our MPS wavefunction reproduces this physical picture. It is convenient to work with the X -gauge and use the matrix in the eigen basis of Pauli- Z . We also assume $c < 1$ and adopt the following parametrization

$$A^\sigma = e^{\rho(\sin \theta X + (-1)^\sigma \cos \theta Z)}$$

Let L denote the total length of the system, then the weight of a single domain wall of length x is

$$\frac{\text{Tr}((A^0)^{L-x}(A^1)^x)}{\text{Tr}(A^0)^L} \approx \frac{1 - \cos 2\theta}{2} + \frac{1 + \cos 2\theta}{2} e^{-2\rho x}, \quad L \gg x > 1 \quad (71)$$

Thus, only when $\theta = 0$, the probability is exponentially small in the length of the domain wall and we have a ferromagnetic phase. This is also consistent with the eigenvalue analysis. Furthermore, the line tension of the domain wall is $\rho = \tanh^{-1} c$.

- In the paramagnetic phase, it is more convenient to describes the physics using $\mathbb{Z}_2$ charge. The wavefunction is required to be $\mathbb{Z}_2$ symmetric and can only contain pairs of $\mathbb{Z}_2$ charge. Not only the pair creation but also the propagation of each $\mathbb{Z}_2$ charge is suppressed. Namely, the weight of a pair of separated $\mathbb{Z}_2$ charge should be exponentially small in their distance. Now, let us reproduce this picture using our MPS wavefunction. It is convenient to work with the Z -gauge and use the matrix in the eigen basis of Pauli- X . We use the following parametrization (assuming $c < 1$)

$$A^+ = \sqrt{2}e^{\alpha Z}, \quad A^- = \sqrt{2}\beta X$$

Let L denote the total system size and x the separation of a single pair of charge, the weight of such a configuration is

$$\frac{\text{Tr}((A^+)^{L-x-2}A^-(A^+)^xA^-)}{\text{Tr}(A^+)^L} \approx \beta^2 e^{-2\alpha(x-1)} \quad (72)$$

As long as $\alpha \neq 0$, the weight is exponentially small in x and the wavefunction is in the paramagnetic phase. This is also consistent with the eigenvalue analysis. $\beta = c \sin \theta$ is probability of creating a $\mathbb{Z}_2$ charge, $\alpha = \tanh^{-1}(c \cos \theta)$ is the virtual energy cost for the charge propagation.

4.3 Mean-field analysis

In this subsection, we finish our mean-field analysis. We will obtain the phase diagram by optimizing the energy for a given J/h and plot the order parameter. We will see that the MPS variational wavefunction exhibits a *first-order* transition at $J/h = \frac{1}{2}(1 + \sqrt{2})$.

The expectation value of the Hamiltonian is

$$\langle H \rangle_\Psi = \frac{\langle \Psi | H | \Psi \rangle}{\langle \Psi | \Psi \rangle} = - \sum_i J \frac{\langle \Psi | Z_i Z_{i+1} | \Psi \rangle}{\langle \Psi | \Psi \rangle} + h \frac{\langle \Psi | X_i | \Psi \rangle}{\langle \Psi | \Psi \rangle} \quad (73)$$

And the order parameter is defined to be the square root of the two-point function of Pauli-Z

$$m \equiv \sqrt{\langle Z_0 Z_{L/2} \rangle_\Psi} = \sqrt{\frac{\langle \Psi | Z_0 Z_{L/2} | \Psi \rangle}{\langle \Psi | \Psi \rangle}} \quad (74)$$

The general analysis in Sec.2.2 tells us that, when the double tensor has a unique largest eigenvalue, these expressions take simple forms in the thermodynamic limit. The one-point function of a local operator can be simplified to (30) and thus the Hamiltonian density can be written as⁸

$$\lim_{L \rightarrow \infty} \frac{\langle H \rangle_\Psi}{L} = \begin{cases} \frac{1}{\lambda_1^2} (-J \langle v_1 | E[Z_1 Z_2] | v_1 \rangle - h \lambda_1 \langle v_1 | E[X] | v_1 \rangle) & \lambda_1 > \lambda_2 \\ \frac{2}{\lambda_1^2} (-J \langle v_1 | E[Z_1 Z_2] | v_1 \rangle - h \lambda_1 \langle v_1 | E[X] | v_1 \rangle) & \lambda_1 = \lambda_2 \end{cases} \quad (75)$$

The two point function is simplified to (32) and thus the order parameter can be written as⁹

$$m = \begin{cases} 0 & \lambda_1 > \lambda_2 \\ \frac{1}{\lambda_1} | \langle v_1 | E[Z] | v_2 \rangle | & \lambda_1 = \lambda_2 \end{cases} \quad (76)$$

For generic value of J/h , the energy density in the parameter space and the corresponding saddle point is shown in Fig. 2. For small J/h , the saddle point is close to the corner $c = 1, \theta = \pi/2$. As we increase J/h , the saddle point gradually moves downward and lands on the real axis when $J/h \approx 1.2$, which is shown in Fig. 3 (a). At the same time, the order parameter jumps to a nonzero value, which is shown in Fig. 3 (b). Therefore, in contrary to the ordinary mean-field result, the MPS mean-field theory yields a first-order transition!

The transition point can also be determined exactly. Let us focus on the vicinity of the transition point and fix $h = 1$ for simplicity. Then, θ is small, which allows us to do a Taylor expansion of the energy density in θ . Plugging the expression of eigenvalue and eigenvector into (75) and performing the Taylor expansion, we find that

$$E(J, c, \theta) \approx E_0(J, c) + E_2(J, c) \theta^2 + E_4(J, c) \theta^4 + \mathcal{O}(\theta^6) \quad (77)$$

Let us inspect each term:

⁸To derive the expression for $\lambda_1 = \lambda_2$ ($\theta = 0$), just notice that $\langle v_1 | E[Z_1 Z_2] | v_1 \rangle = \langle v_2 | E[Z_1 Z_2] | v_2 \rangle$ and $\langle v_1 | E[X] | v_1 \rangle = \langle v_2 | E[X] | v_2 \rangle$. Thus, we can simply extend the expression for $\lambda_1 > \lambda_2$ to the degenerate case by adding a factor of 2.

⁹There are a few intermediate steps to derive the expression for $\lambda_1 = \lambda_2$. First, we need to write down the analytical expression of the eigenvectors and then one can find that $\langle v_i | E[Z] | v_i \rangle = 0$ for all $i = 1, \dots, 4$. Thus, only the off-diagonal elements $\langle v_i | E[Z] | v_j \rangle, i \neq j$ can be nonzero. The leading contribution comes from the two-fold degenerate vectors with the largest eigenvalue. Further using the fact that $E[Z]$ and $v_{i=1,2}$ are real, we get the expression.

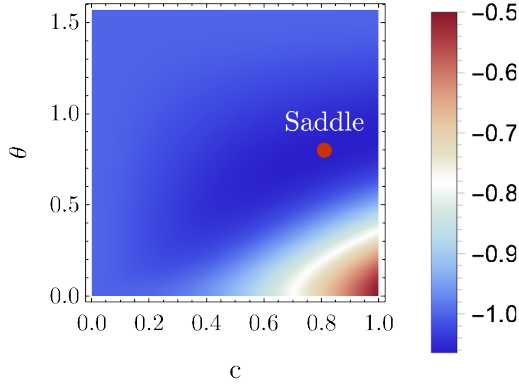


Figure 2: Energy density in the parameter space and the saddle point. In this plot, we choose $J/h = 0.5$, which is the critical point in the ordinary mean-field theory. Using the MPS mean-field ansatz, the saddle point is above the real axis and does not exhibit the symmetry breaking order.

- The first term is independent of θ and is what the ordinary mean-field theory gives

$$E_0(J, c) = \frac{-1 + c^4 - 4c^2 J}{(1 + c^2)^2} \quad (78)$$

Optimization with respect to c for a given J yields a transition point at $J = 1/2$, which is exactly the ordinary mean field result. When $J > 1/2$, the minimal of $E_0(J, c)$ is at

$$c_{min}^{(0)} = \sqrt{\frac{2J - 1}{2J + 1}} \quad (79)$$

- Now, let us consider the quadratic term

$$E_2(J, c) = -\frac{2(2c^4 J - 8c^2 J + c^6 + 6c^4 + 7c^2 - 2J + 2)}{(c^2 + 1)^3} \quad (80)$$

One can find $E_2(J, c) < 0$ for $0 < J < 1$ and therefore the wavefunction can lower its energy by having a nonzero θ , i.e. no symmetry breaking. The quartic term is always positive and forbids θ from being too large.

When J exceeds 1, $E_2(J, c)$ starts to have a zero $c_{zero}^{(2)}$ on the positive c axis. When $c_{zero}^{(2)} \geq c_{min}^{(0)}$, the system does not have to lower its energy by having a nonzero θ . Thus, the transition point is determined by

$$c_{zero}^{(2)} = c_{min}^{(0)} \Rightarrow J_{transition} = \frac{1}{2}(\sqrt{2} + 1) \quad (81)$$

After the transition point, the minimal of $E_0(J, c)$ becomes the global minimal and we have

$$c_{min} = \sqrt{\frac{2J - 1}{2J + 1}}, \quad J > \frac{1}{2}(\sqrt{2} + 1) \quad (82)$$

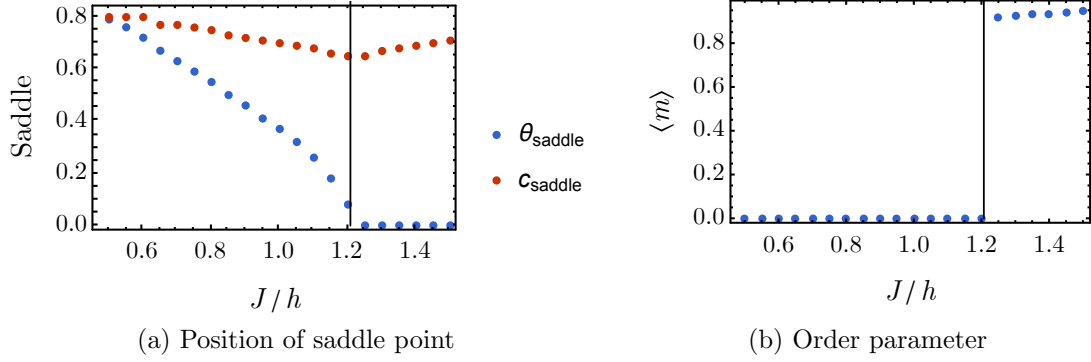


Figure 3: Position of saddle point and order parameter. The transition point is $J/h \approx 1.2$. The order parameter shows a discontinuity, which implies that the transition is first-order.

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